

Limits Graphically and Numerically

ANSWER KEY



Estimating limits from a table

- 1 A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \rightarrow 3} g(x)$.
Both sides approach 7: $\lim_{x \rightarrow 3} g(x) = 7$.
- 2 Using a table of values with x near 0 (radians), estimate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).
Values approach 0.50 from both sides, consistent with the exact limit $\frac{1}{2}$.

Reading limits from a graph

- 3 The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ ($x \neq 1$), state $\lim_{x \rightarrow 1} f(x)$.
 $\lim_{x \rightarrow 1} f(x) = 2$ (even though $f(1)$ itself is undefined — a removable discontinuity).
- 4 A function h has $\lim_{x \rightarrow 2^-} h(x) = 5$ and $\lim_{x \rightarrow 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \rightarrow 2} h(x)$ and explain whether h is continuous at $x = 2$.
 $\lim_{x \rightarrow 2} h(x) = 5$, since both one-sided limits agree. But h is NOT continuous at $x = 2$, since $\lim_{x \rightarrow 2} h(x) = 5 \neq h(2) = 8$ — this is a removable discontinuity.

One-sided limits and limits that fail to exist

- 5 For $f(x) = \frac{x}{|x|}$, find:
a $\lim_{x \rightarrow 0^+} f(x) = 1$ **b** $\lim_{x \rightarrow 0^-} f(x) = -1$ **c** $\lim_{x \rightarrow 0} f(x)$ does not exist
- 6 A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, and state whether $\lim_{x \rightarrow 1} f(x)$ exists.
 $\lim_{x \rightarrow 1^-} f(x) = 1^2 = 1$. $\lim_{x \rightarrow 1^+} f(x) = 3 - 1 = 2$. Since the one-sided limits disagree ($1 \neq 2$), $\lim_{x \rightarrow 1} f(x)$ does not exist.

Algebraic evaluation

7 Evaluate each limit by direct substitution or factoring:

a $\lim_{x \rightarrow 4} (3x^2 - 5x + 1) = 3(16) - 20 + 1 = 29$

b $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$

c $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2} = \lim_{x \rightarrow -2} (x + 3) = 1$

8 Evaluate $\lim_{x \rightarrow 6} \frac{\sqrt{x+9} - 3}{x}$ by rationalizing the numerator.

Multiply by $\frac{\sqrt{x+9} + 3}{\sqrt{x+9} + 3} \cdot \frac{x}{x(\sqrt{x+9} + 3)} = \frac{1}{\sqrt{x+9} + 3} \rightarrow \frac{1}{6}$.